Ha Bac Ky

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Objective

Robotics & Control Engineering student seeking an internship in robotics, autonomous systems, or motion control. Experienced in dynamic modeling, nonlinear control design, and embedded implementation. Passionate about bridging theory and real-world robotic applications.

Education

Hanoi University of Science and Technology (HUST) Hanoi, Vietnam
B.Eng. in Control and Automation Engineering *Oct 2022 – Expected Oct 2026*

- Relevant Coursework: Robotics & Control Systems, Dynamic System Modeling, Optimal Control

Research Experience

MEG Lab – MOCAR Laboratory HUST, Hanoi
Undergraduate Research Assistant *2024 – Present*

- Conduct research on robotics dynamics and control algorithms for legged and wheeled robotic systems.
- Study and implement whole-body control (WBC) methods for legged robots, including floating-base dynamics formulation.
- Develop and validate dynamic models for robotic systems using MATLAB/Simulink.
- Collaborate on laboratory research projects in motion planning and nonlinear control.

Projects

Model Predictive Control for Skid-Steering Mobile Robot 2025 – Present

- Studying dynamic modeling of skid-steering mobile robots, including wheel-terrain interaction and slip effects.
- Investigating Model Predictive Control (MPC) for trajectory tracking, focusing on handling input constraints and nonlinear dynamics.
- Exploring integration of MPC with advanced control strategies to improve robustness and real-time performance.

Dynamic Modeling of a Jumping-Capable Bipedal Robot (Ascento) 2024

- Derived floating-base dynamic model for a wheel-legged robot using Euler-Lagrange formulation.
- Analyzed leg kinematics and formulating whole-body control (WBC) with contact constraints.

Trajectory Tracking Control of a Differential-Drive Robot 2025

- Developed kinematic model and implemented Pure Pursuit trajectory tracking algorithm.
- Deployed controller on STM32 microcontroller in C; achieved stable tracking on curved reference paths.
- Tuned control parameters to minimize cross-track error during hardware testing.

Motion–Force Control of Heavy Load Lifting Using Collaborative Robots 2025

- Studied centralized control strategies for multi-robot collaborative manipulation tasks.
- Analyzed internal interaction forces during coordinated heavy-load lifting to ensure safe operation.

Temperature Control System for Mini Oven 2025

- Designed and implemented a closed-loop PID temperature control system on STM32.
- Achieved stable temperature regulation with steady-state error within $\pm 1^\circ$C.

Publications

Hybrid Backstepping and Fast Terminal Sliding Mode Control Accepted

- Hoang-Tung Nguyen, Duc-Manh Nguyen, **Bac-Ky Ha**, Thi-Van-Anh Nguyen, Manh-Linh Nguyen. *"Hybrid Backstepping and Fast Terminal Sliding Mode Control for Balancing of Two-Wheeled Mobile Robots."* Under review, **JST: Smart Systems and Devices**.

Technical Skills

Programming Languages: MATLAB/Simulink, C/C++ (basic)

Control Theory: PID Control, LQR, Sliding Mode Control (SMC), Backstepping, Whole-Body Control, Dynamic Modeling, Robot Kinematics

Tools & Platforms: MATLAB/Simulink, STM32 (Embedded Programming)

Languages: Vietnamese (native), English – TOEIC (Institutional Test – HUST): 715

Research Interests

Robotics Control · Whole-Body Control · Legged & Wheeled Robot Dynamics · Skid-Steering Mobile Robots · Nonlinear Control Systems